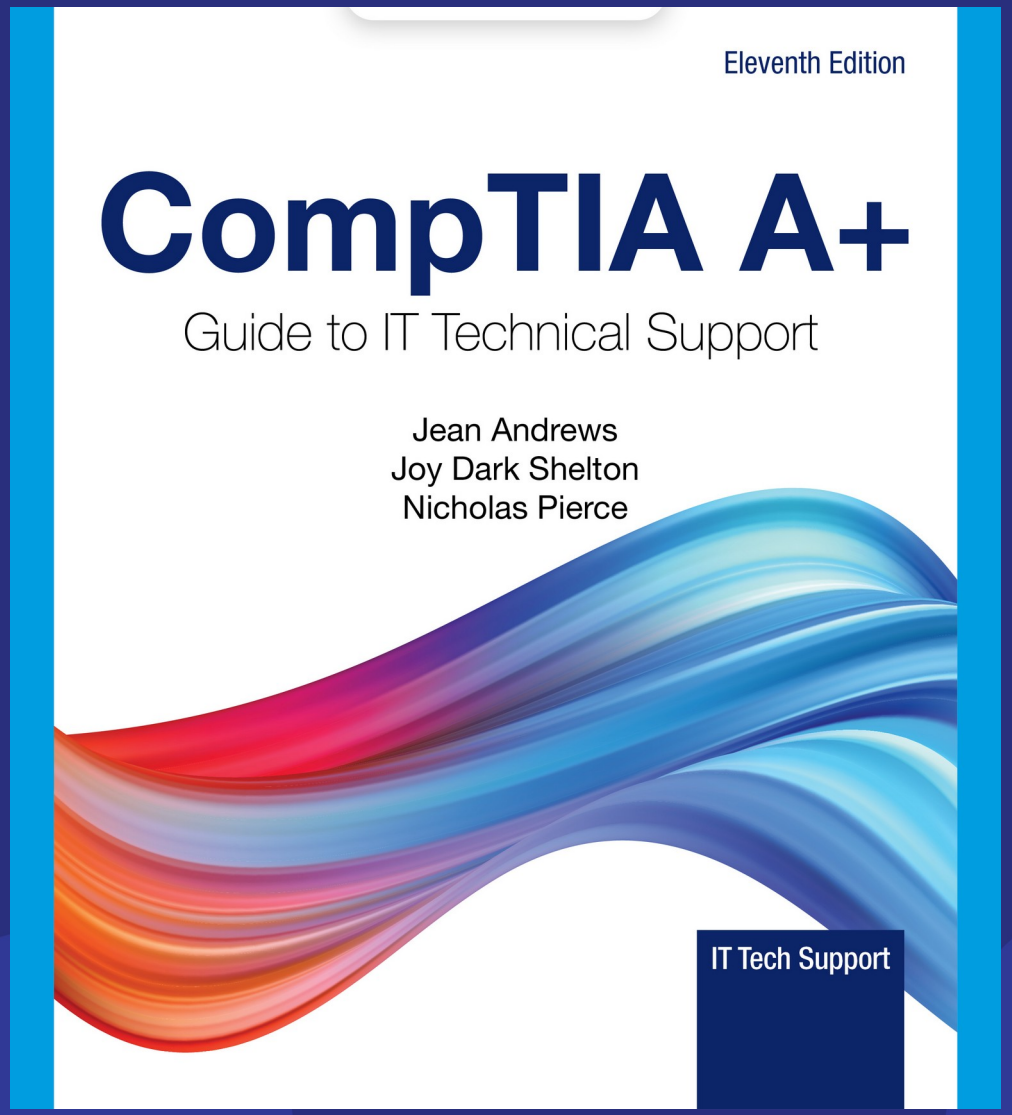

Module 1

Taking a Computer Apart and Putting It Back Together

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Part 1

CompTIA A+ Core 1 (220-1101)

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Icebreaker: Interview Simulation

1. The class will be broken up into pairs of students.
2. Each pair of students will interview each other to discover interesting or unusual facts.
3. Then each pair will introduce each other to the class.
4. Think about connecting a story from your personal experience to topics that are relevant to this course

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Module Objectives

- By the end of this module, you should be able to:
- Disassemble and reassemble a desktop computer safely, and identify external ports and major components inside a desktop. Describe how they connect and are compatible. Identify various tools you need as a computer hardware technician.
- Disassemble and reassemble a laptop computer safely, and identify various external ports and slots and major internal components. Understand special concerns when supporting and maintaining laptops.

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Exploring a Desktop Computer

- Some guidelines for protecting yourself and the equipment include the following:
  - Remove loose jewelry
  - Watch out for sharp edges on computer cases
  - Never remove the cover or put your hands inside the monitor or power supply
  - Power down the system and unplug it
  - Press and hold the power button for three seconds to completely drain power
  - Never touch the inside of a computer that is turned on
  - Always wear an ESD strap or use a ground mat if an ESD strap is not available
  - Place equipment in an antistatic bag

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Step 1: Planning and Organizing Your Work and Gathering Your Tools

- Make notes for backtracking
- Stay organized by keeping small parts in one place
- Some essential tools used by a computer hardware technician include:
  - ESD strap (ground bracelet)
  - Flat-head screwdriver
  - Phillips-head or cross-head screwdriver
  - Torx screwdriver set (size T15)
  - Insulated tweezers

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Step 2: Opening the Case (1 of 9)

- A computer case for any type of computer is sometimes called the **chassis**
  - It houses the power supply, motherboard, processor, memory modules, expansion cards, hard drive, optical drive, and other drives
  - It can be a **tower case**, **desktop case**, all-in-one case, or a mobile case
- **Loopback plugs** are used to test a network port in a computer or other device to make sure the port is working
  - May also be used to test the throughput or speed of a port

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Step 2: Opening the Case (2 of 9)

- Back up important data
- Power down the system and unplug it along with other peripherals as well
- Press and hold the power button for about 3 seconds
- Have a plastic bag or cup handy to hold screws
- Open the case cover
- Clip your ESD strap to the side of the computer case

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Step 2: Opening the Case (3 of 9)

Figure 1-14 Inside the computer case

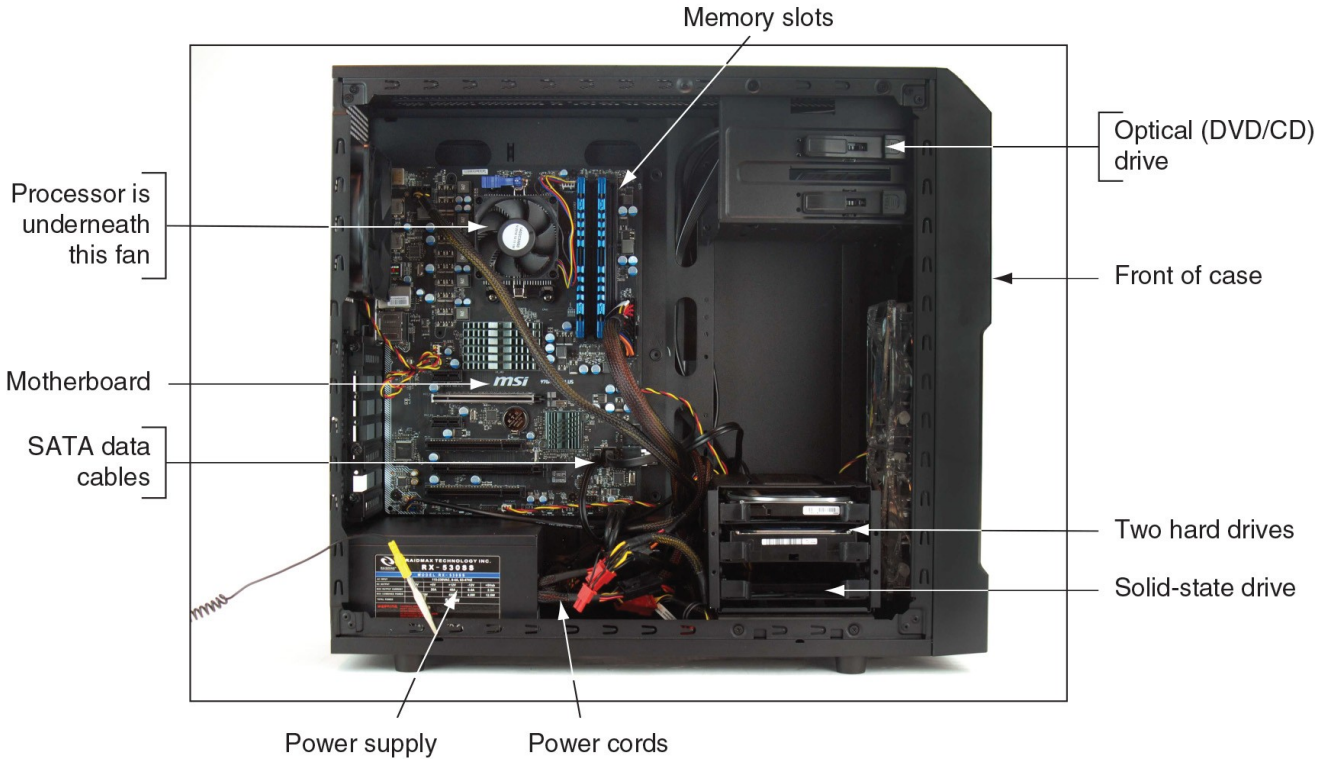

Figure 1-14 Inside the computer case

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Step 2: Opening the Case (4 of 9)

- Main components installed in the case include the following:
  - The motherboard, processor, and cooler
  - Expansion cards
  - Memory modules
  - Hard drives and other drives
  - The power supply

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Step 2: Opening the Case (5 of 9)

Figure 1-18 A DIMM holds RAM and is mounted directly on a motherboard

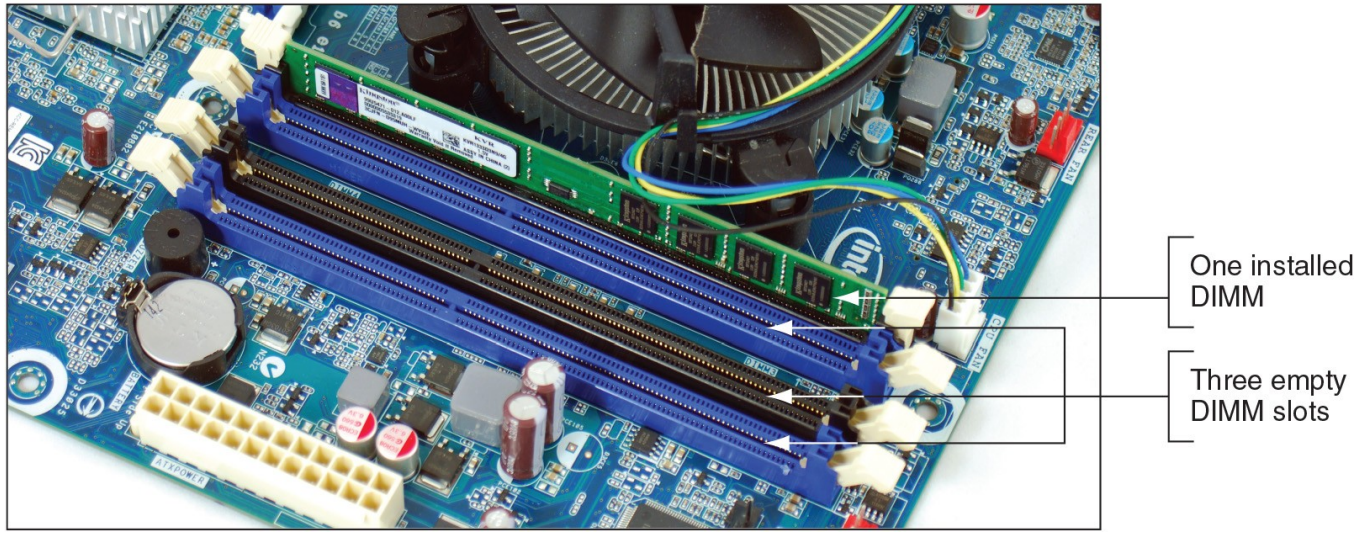

Figure 1-18 A DIMM holds RAM and is mounted directly on a motherboard

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Step 2: Opening the Case (6 of 9)

- Form Factors Used by Desktop Cases, Power Supplies, and Motherboards
  - Standards that describe the size, shape, screw hole positions, and major features of these interconnected components are called **form factors**
  - Form factors are necessary so that all will be compatible with each other
- Two form factors used by most desktop and tower computer cases and power supplies include ATX (Advanced Technology Extended) and microATX

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Step 2: Opening the Case (7 of 9)

- An ATX power supply has a variety of power connectors including the following:
  - 4-pin and 8-pin auxiliary connectors are used to provide an additional 12 V of power for evolving CPUs
  - 24-pin or 20+4-pin P1 connector
    - The older 20-pin P1 connector still worked in this connector
    - Supported the new **PCI Express (PCIe)** slots
  - 6-pin and 8-pin PCIe connectors connect directly to the video card
    - Video cards draw the most power in a system

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Step 2: Opening the Case (8 of 9)

Figure 1-27 This PCIe ×16 video card has a 6-pin PCIe power connector to receive extra power from the power supply

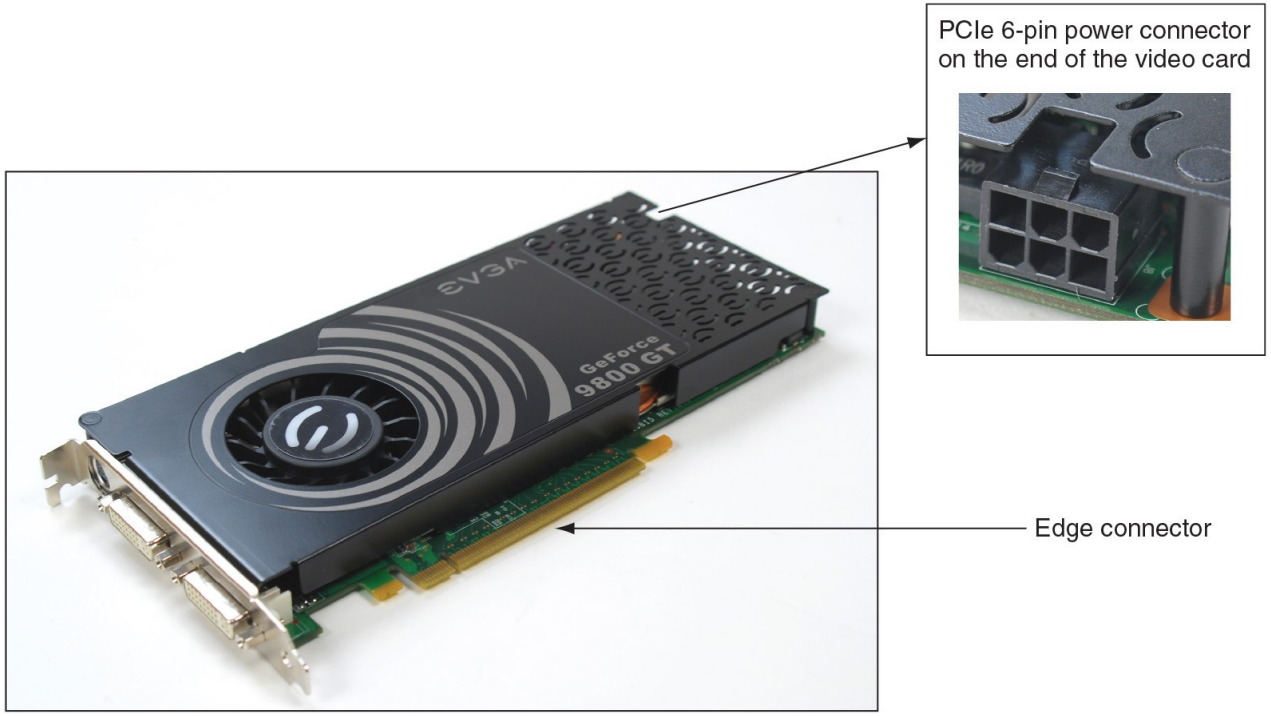

Figure 1-27 This PCIe x 16 video card has a 6-pin PCIe power connector to receive extra power from the power supply

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Step 2: Opening the Case (9 of 9)

- The microATX (mATX) form factor is a major variation of ATX
  - It reduces the total cost of a system by implementing the following:
    - Reduces the number of expansion slots on motherboard
    - Reduces the power supplied to the board
    - Allows for a smaller case size
  - The microATX form factor uses a 24-pin P1 connector
    - It is not likely to have as many extra wires and connectors as those on the ATX power supply

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Step 3: Removing Expansion Cards

- If removing components, draw a diagram of all cable connections to the motherboard, expansion cards, and drives
- To remove expansion cards, do the following:
  - Remove any wire or cable connected to the card
  - Remove the screw holding the card to the case
  - Grasp the card with both hands and remove it by lifting straight up (you can also very slightly rock the card from end to end)
  - Don’t put your fingers on the edge connectors or touch a chip

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Step 4: Removing the Motherboard (1 of 2)

- Depending on the system, you may have to remove the drives and/or power supply to get to the motherboard
- To remove the motherboard, do the following:
  - Unplug the power supply lines
  - Unplug SATA cables connected to the motherboard
  - Disconnect wires leading from the front of the computer case to the motherboard (called **front panel connectors**)
  - Disconnect any other cables or wires connected to the motherboard
  - Remove the screws that hold the motherboard to the case

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Step 4: Removing the Motherboard (2 of 2)

- **Post Diagnostic Cards** can help discover and report computer errors and conflicts at power-on self test (POST)
  - **Firmware** are programs and data stored on the motherboard
  - **POST** is a series of tests performed by the startup BIOS/UEFI
- If you have a problem that prevents the computer from booting and you suspect it is related to hardware, do the following:
  - Install the POST card in an expansion slot on the motherboard
    - The card monitors the boot process and reports errors

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Step 5: Removing the Power Supply (1 of 3)

- To remove the power supply from the case, do the following:
  - Look for screws that attach the power supply to the computer case
    - Do not remove screws that hold power supply housing together
  - Sometimes power supplies are also attached to the case on the underside by recessed slots
    - Turn the case over and look for slots
    - If present, determine in which direction you need to slide the power supply to free it from the case

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Step 5: Removing the Power Supply (2 of 3)

Figure 1-39 Use a power supply tester to test the output of each power connector on a power supply

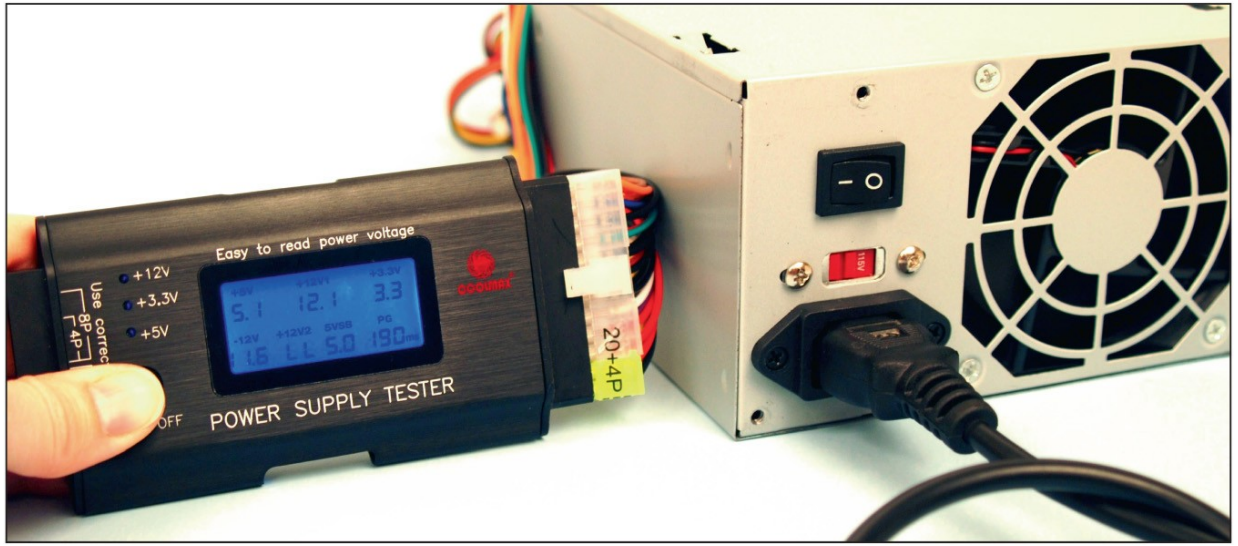

Figure 1-39 Use a power supply tester to test the output of each power connector on a power supply

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Step 5: Removing the Power Supply (3 of 3)

Figure 1-40 This digital multimeter can be set to measure voltage, resistance, or continuity

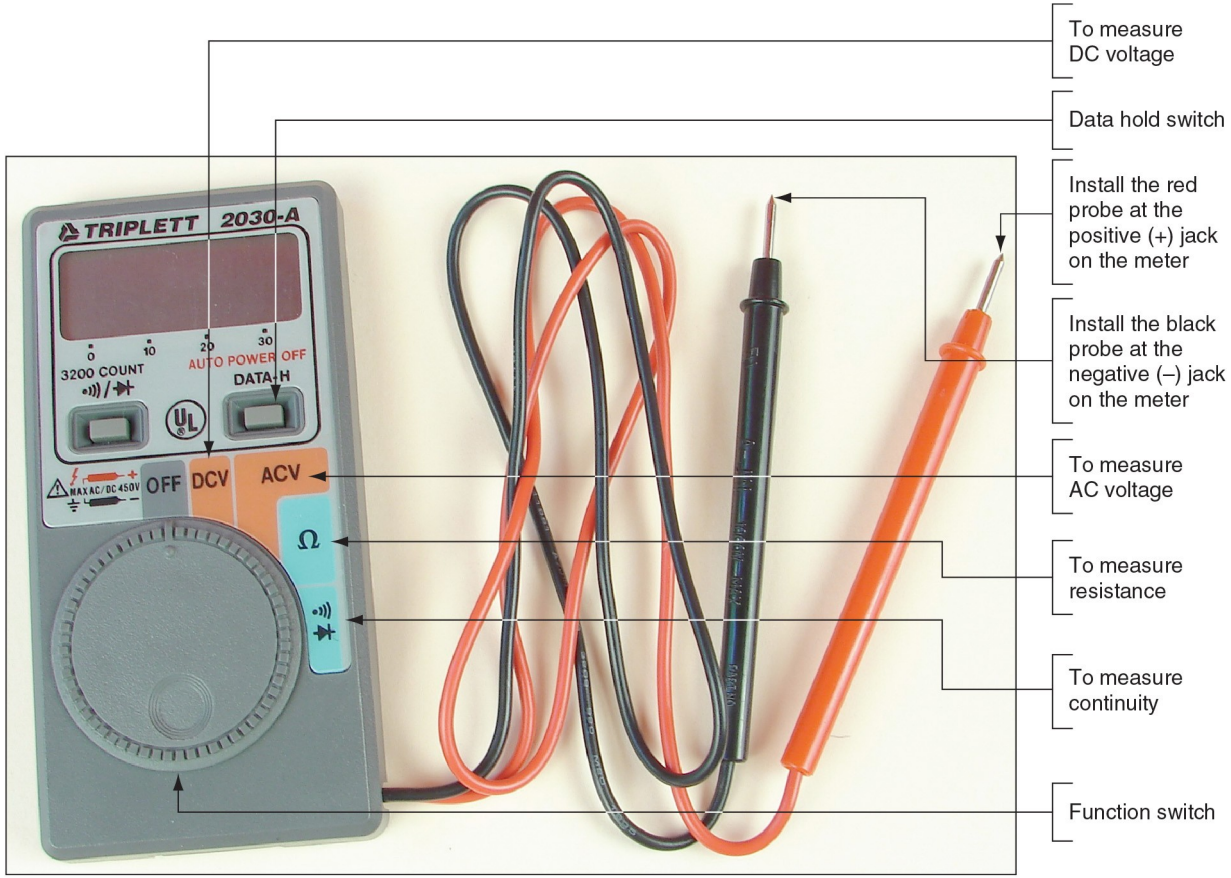

Figure 1-40 This digital multimeter can be set to measure voltage, resistance, or continuity

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Step 6: Removing the Drives (1 of 2)

- A drive receives power by a power cable from the power supply and communicates instructions and data through a cable attached to the motherboard
- Tips to remove drives:
  - Look for screws on each side of the drive attaching the drive to the drive bay
  - There might be a catch underneath the drive
  - Some drive bays have a clipping mechanism to hold the drive in the bay
  - Some drives must be removed through the front of the case
  - Some cases have a removable bay for smaller hard drives

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Step 6: Removing the Drives (2 of 2)

Figure 1-43 To remove this optical drive, first release the clip to release the drive from the bay

Figure 1-43 To remove this optical drive, first release the clip to release the drive from the bay

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Steps to Put a Computer Back Together (1 of 4)

- Refer to any diagrams created during the disassembling process
- 1. Install components in this order: power supply, drives, motherboard, and cards
  - When installing drives, it may be easier to connect cables to the drives before sliding them into the bay
- 2. Place the motherboard inside the case
  - Make sure ports and screw holes are lined up
- 3. Connect the power cords from the power supply to the motherboard

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Steps to Put a Computer Back Together (2 of 4)

- 4. Connect wire leads from the front panel of the case to the front panel header on the motherboard
  - To help orient the larger connectors, look for a small triangle embedded on the connector that marks one of the outside wires as pin 1
    - Line up pin 1 on connector with pin 1 marked on motherboard
- 5. Connect wires to the ports on the front panel of the case
  - Audio ports or USB ports are examples

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Steps to Put a Computer Back Together (3 of 4)

Figure 1-55 Connectors for front panel ports

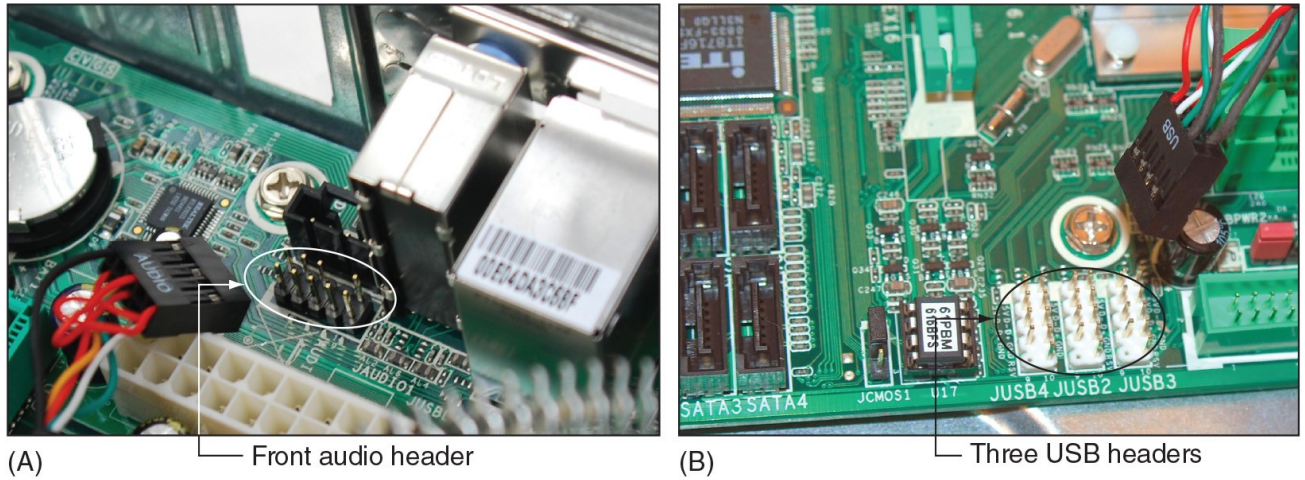

Figure 1-55 Connectors for front panel ports

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Steps to Put a Computer Back Together (4 of 4)

- 6. Install the video card and other expansion cards
- 7. Double-check each connection
- 8. Plug in keyboard, monitor, and mouse
- 9. In a classroom environment, have the instructor check your work
- 10. Turn on the power and check that the PC is working properly
  - If not, it is most likely a loose connection

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Knowledge Check Activity 1-1

- You disassemble and reassemble a desktop computer. When you first turn it on, you see no lights and hear no sounds. Nothing appears on the monitor screen. What is the most likely cause of the problem?
  a. A memory module is not seated properly in a memory slot.
  b. You forgot to plug in the monitor’s external power cord.
  c. A wire in the case is obstructing a fan.
  d. Power cords to the motherboard are not connected.

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Knowledge Check Activity 1-1: Answer

- You disassemble and reassemble a desktop computer. When you first turn it on, you see no lights and hear no sounds. Nothing appears on the monitor screen. What is the most likely cause of the problem?

- **Answer: d. Power cords to the motherboard are not connected.**
- **All the other answers would still allow the system to start the boot even though it might fail. If the motherboard is not getting power, it will not start the boot.**

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First Look at Laptop Computers (1 of 3)

- A **laptop (notebook)** is designed for portability and can be just as powerful as a desktop computer
- Varieties of laptops include:
  - Those with a touch screen that allows you to handwrite on it with a stylus
  - Some are outfitted with a rotating or removable screen
  - A **netbook** is a smaller and less expensive with fewer features
  - **All-in-one computer** has the monitor and computer case built together and uses components common to a laptop and a desktop

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First Look at Laptop Computers (2 of 3)

Figure 1-56 A laptop, netbook, and all-in-one computer

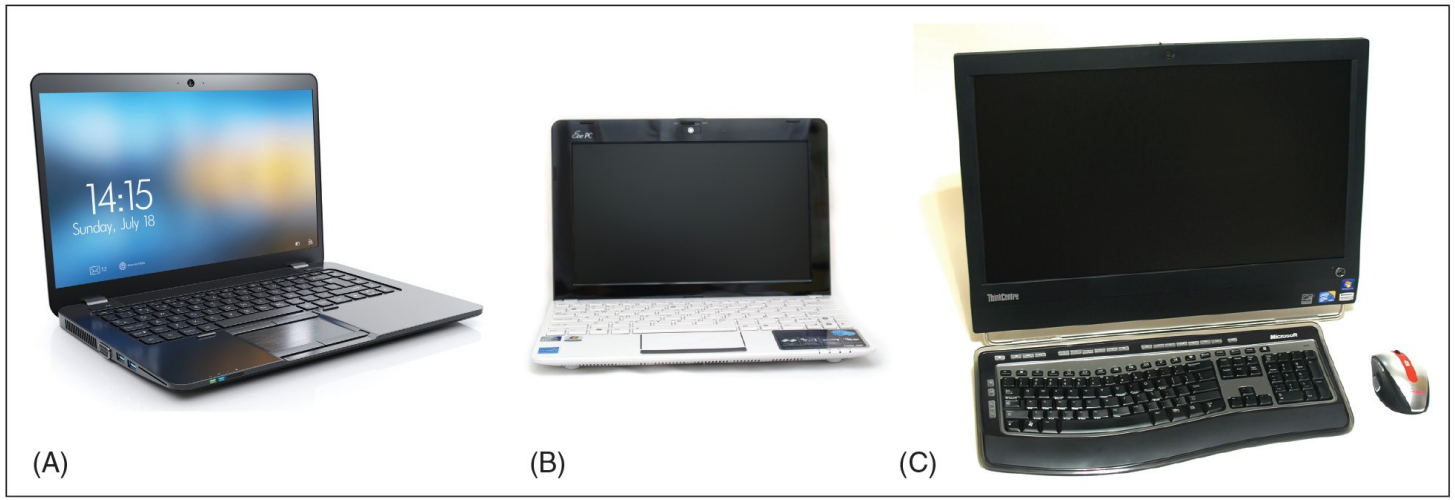

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Figure 1-56 A laptop, netbook, and all-in-one computer

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First Look at Laptop Computers (3 of 3)

- Ports common to laptops include USB, network, and audio ports
- When a laptop is missing a port or slot, you can use a USB dongle to provide the port or slot
  - Examples include the following:
    - **USB to RJ-45 dongle** to connect to a wired network
    - **USB to Wi-Fi dongle** to connect to a wireless network
    - **USB to Bluetooth** adapter to connect to a Bluetooth wireless device

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Docking Stations and Port Replicators (1 of 2)

- Some laptops have a connector called a docking port that connects to a docking station
- A **docking station** provides ports to allow a laptop to easily connect to a full-sized monitor, keyboard, AC power adapter, and other peripheral devices
- A **port replicator** (universal docking station) is a device that provides ports to allow a laptop to easily connect to peripheral devices
- To use a docking station or port replicator plug all peripherals into the docking station or port replicator, then connect the laptop to the station

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Docking Stations and Port Replicators (2 of 2)

Figure 1-63 A docking station for a Lenovo ThinkPad

Source: Lenovo

Figure 1-63 A docking station for a Lenovo ThinkPad

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Special Considerations When Supporting Laptops (1 of 4)

- Laptops and their replacement parts cost more than desktop PCs
- Factors to consider that apply more to laptop than desktop computers:
  - The original equipment manufacturer (OEM) warranty
  - The service manuals and diagnostic software provided by the manufacturer
  - The customized installation of the OS that is unique to laptops
  - The need to order replacement parts directly from the laptop manufacturer or authorized source

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Special Considerations When Supporting Laptops (2 of 4)

- Warranty Concerns
  - Always check to see if the laptop is under warranty before servicing
  - When contacting technical support have the following available:
    - Laptop model and serial number, purchaser name, phone number, address
- Service Manuals and Other Sources of Information
  - Service manuals save time
    - They enable safe laptop disassembly

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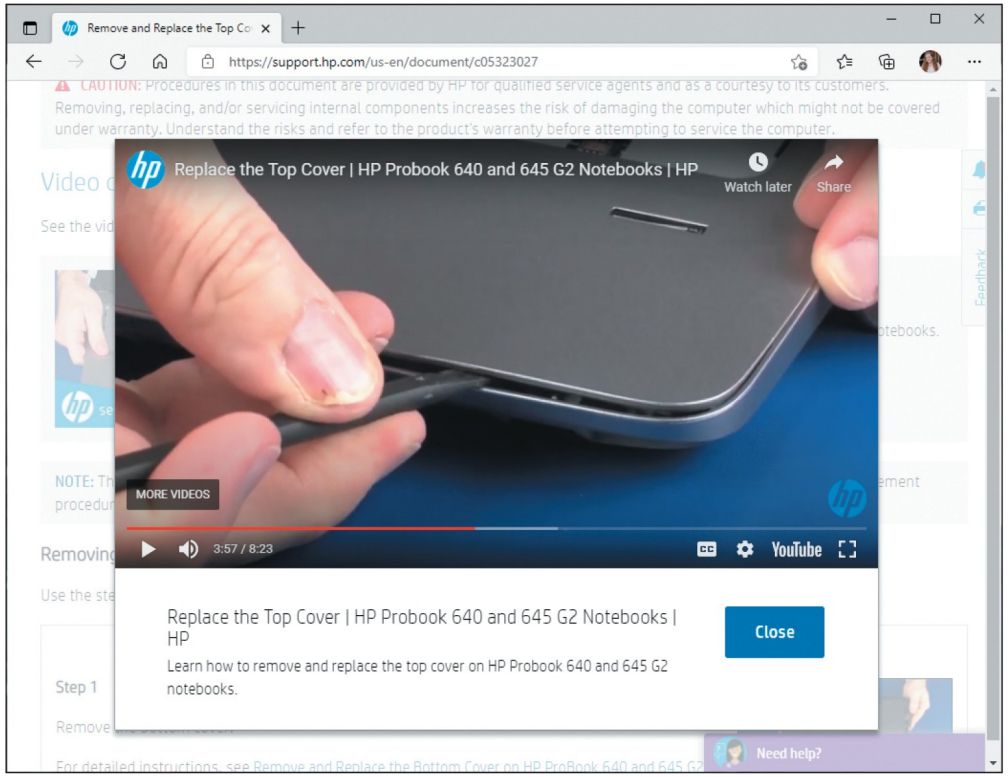
Special Considerations When Supporting Laptops (3 of 4)

Figure 1-67 The HP website (*support.hp.com*) provides detailed instructions and videos for troubleshooting and replacing components



Source: hp.com

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Special Considerations When Supporting Laptops (4 of 4)

- Most laptop manufacturers provide diagnostic software that can help you test components to determine which component needs replacing
  - Sources include, manufacturer's website, CDs bundled with the notebook, or on the hard drive itself

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Working Inside a Laptop Computer (1 of 4)

- It may become necessary to open a laptop case to upgrade memory, exchange a hard drive, or replace a failed component
- Screws and nuts on a laptop are smaller than a desktop, requiring smaller tools
- Working on laptops requires extra patience
- Always wear an ESD strap
- Tape screws on a piece of paper or beside the manufacturer documentation
  - Work methodically to keep screws and components organized

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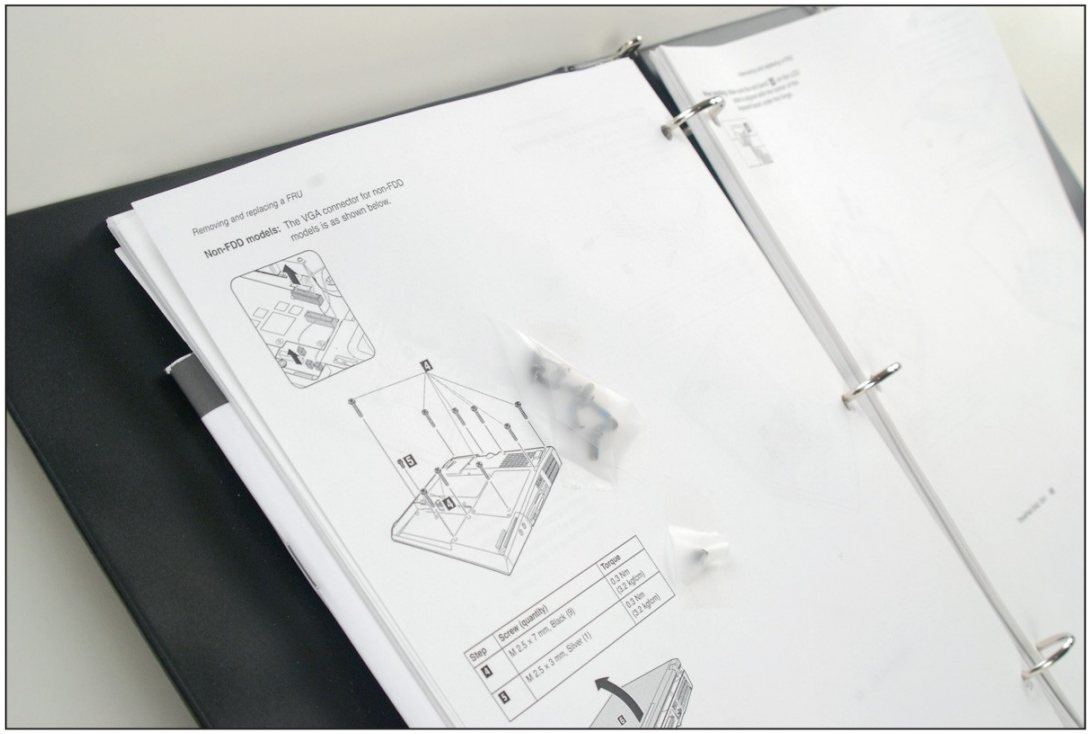
Working Inside a Laptop Computer (2 of 4)

Figure 1-73 Tape screws beside the step in the manufacturer documentation that told you to remove the screw



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Working Inside a Laptop Computer (3 of 4)

- A few tips to consider when disassembling a laptop include the following:
  - Find the hardware service manual
  - Opening the case might void the warranty
  - Take your time and do not force anything
  - Pry up plastic covers with dental pick or screwdriver
  - Plastic screws may be used only once
  - Disassemble components in order
  - Use a plastic or metal spudger to slide along the seal and pry open the case

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Working Inside a Laptop Computer (4 of 4)

Figure 1-74 To disconnect a ZIF connector, first lift up on the lever or locking bar to release the latch, and then remove the cable using the pull tab, which is blue on this laptop

Figure 1-74 To disconnect a ZIF connector, first lift up on the lever or locking bar to release the latch, and then remove the cable using the pull tab, which is blue on this laptop

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Exploring Laptop Internal Components

- A list of important components and the typical order you remove them:
  - 1. Remove or disable the battery pack
  - 2. Remove the hard drive
  - 3. Remove memory
  - 4. Remove the wireless card
  - 5. Remove the optical drive
  - 6. Crack the case
  - 7. Remove the keyboard bezel

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Exploring Inside an All-in-One Computer (1 of 2)

- An all-in-one computer uses a mix of components sized for a desktop and a laptop
  - For some components, you will need to buy replacements from the manufacturer because they are most likely proprietary
  - See the service manual for specific directions about replacing parts

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Exploring Inside an All-in-One Computer (2 of 2)

Figure 1-84 Components inside an all-in-one computer

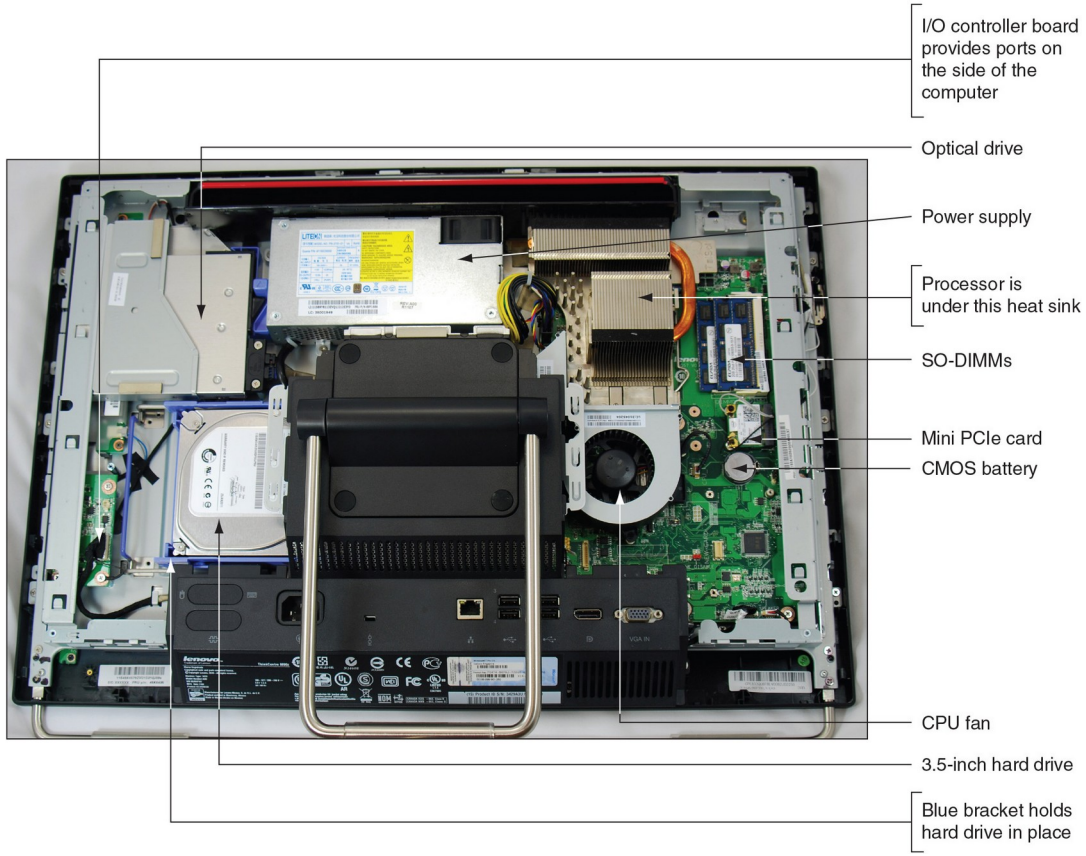

Figure 1-84 Components inside an all-in-one computer

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Maintaining Laptops

- A few general guidelines to follow when maintaining laptops:
  - Do not touch the LCD panel with sharp objects
  - Do not pick up or hold a laptop by the lid
  - Use only battery packs recommended by the manufacturer
  - Do not tightly pack the laptop in a suitcase – use a carrying case
  - Do not move while hard drive is being accessed
  - Do not put close to appliances generating strong magnetic field
  - Always use passwords to protect your laptop

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Knowledge Check Activity 1-2

- A four-year-old laptop will not boot and presents error messages on-screen. You have verified with the laptop technical support that these error messages indicate the wireless card has failed and needs replacing. What is the first step you should take to prepare for the repair?
  a. Ask yourself if replacing the wireless card will cost more than purchasing a new laptop.
  b. Find a replacement wireless card.
  c. Find the service manual to show you how to replace the wireless card.
  d. Ask if the laptop is still under warranty.

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Knowledge Check Activity 1-2: Answer

- A four-year-old laptop will not boot and presents error messages on-screen. You have verified with the laptop technical support that these error messages indicate the wireless card has failed and needs replacing. What is the first step you should take to prepare for the repair?

- **Answer: d. Ask if the laptop is still under warranty.**
- **Although all the options are good steps to take before attempting the repair, the first step should be to ask is if the laptop is still under warranty.**

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Self Assessment

- Have you ever worked inside a desktop computer or laptop computer?
- What measures did you take in order to keep you and the equipment safe?

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Summary

- Now that the lesson has ended, you should be able to:
- Disassemble and reassemble a desktop computer safely, and identify external ports and major components inside a desktop. Describe how they connect and are compatible. Identify various tools you need as a computer hardware technician.
- Disassemble and reassemble a laptop computer safely, and identify various external ports and slots and major internal components. Understand special concerns when supporting and maintaining laptops.

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